Aspirin and acetaminophen as COX-1 inhibitors

Is acetaminophen as effective as aspirin (acetylsalicylic acid) as an inhibitor of cyclooxygenase-1 (COX-1) catalysis of arachidonic acid metabolism, determined by calculating the IC_{50} values?

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1. Introduction

a. Background

Most individuals are familiar with analgesics: they are commonplace drugs that induce pain relief. Non-steroidal anti-inflammatory drugs (NSAIDs) are a class of analgesics that relieve pain by reducing inflammation in the body. During the Covid-19 pandemic, much work was done to determine the best analgesics to reduce the fevers of Covid-19 patients, and it would be fascinating to explore the chemical reactions that occur when analgesics, such as aspirin and acetaminophen, are ingested to reduce pain.

The fundamental idea behind chemical reactions is that reactant molecules collide with each other with sufficient energy and a specific geometric alignment to form product molecules, known as collision theory. Any step taken to increase the rate of a chemical reaction affects the collision of the reactant molecules in some way. One such way is through catalysis, using a substance that is not consumed by the overall reaction to provide an alternative reaction mechanism that lowers the activation energy, the energy required for molecules to collide (Bjorkov et al.).

Enzymes are biological catalysts, unique among other catalysts because they catalyze the reactions of specific substances, or chemical substrates, to form products. For an enzymatic reaction to occur, an enzyme must bind the substrate to its active site, which facilitates the alignment of the reactants so that the collisions between molecules are focused and require less energy. The active site of an enzyme will only bind certain substrates to

catalyze the reaction, assuring a property of specificity among substrates. When a specific inhibitor seeks to block the catalytic effect of an enzyme, the inhibitor binds itself to the enzyme's active site instead of the substrate, reducing the efficacy of the enzyme to catalyze the reaction (Robinson).

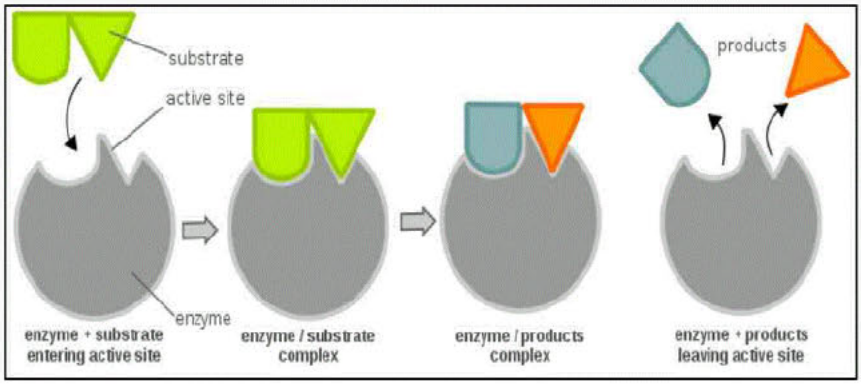


Figure #1: Representation of the enzyme-substrate complex

Cyclooxygenase-1 (or COX-1, EC 1.14.99.1) is one of the isoforms of cyclooxygenase, an oxygenase enzyme that catalyzes the metabolism of arachidonic acid into prostaglandins, a subsection of eicosanoids. COX-1 is unique among most enzymes in that it has two catalytic functions based on two active sites, a serine and a tyrosine active site (Rouzer and Marnett). In the first, the cyclooxygenase phase, an arachidonic acid molecule reacts with two oxygen gas molecules at the serine active site to form Prostaglandin G_2 ($C_{20}H_{32}O_6$), which is unstable at standard conditions.

$$C_{20}H_{32}O_{2\,(aq)} + 2O_{2\,(g)} \xrightarrow{\text{COX-1 (Oxygenase)}} C_{20}H_{32}O_{6\,(aq)}$$

As an unsaturated fatty acid, the four cis-double bonds of arachidonic acid make it readily reactive and prone to oxidation, saturating some of the double bonds (Limonciel).

The second stage is the peroxidase phase at the tyrosine active site, where the unstable Prostaglandin G_2 is reduced to form Prostaglandin H_2 ($C_{20}H_{32}O_5$), which is also unstable, and water. In this phase of metabolism, two electron donors (accompanying two hydrogen cations) reduce the peroxide branch (-OOH) Prostaglandin G_2 into a hydroxyl branch (-OH), leaving the leftover oxygen molecule which forms water molecules with the H^+ ions (Rouzer and Marnett).

$$C_{20}H_{32}O_{6\,(aq)} + 2e^-_{(aq)} + 2H^+_{(aq)} \xrightarrow{\text{COX-1 (Peroxidase)}} C_{20}H_{32}O_{5(aq)} + H_2O_{(l)}$$

After this step, Prostaglandin H_2 can be isomerized to form one of numerous stable prostaglandins, one of which is Prostaglandin E_2 (PGE_2), which is the product that this paper will be referring to as the product of cyclooxygenase-1 metabolism (Lone and Taskén; Rouzer and Marnett).

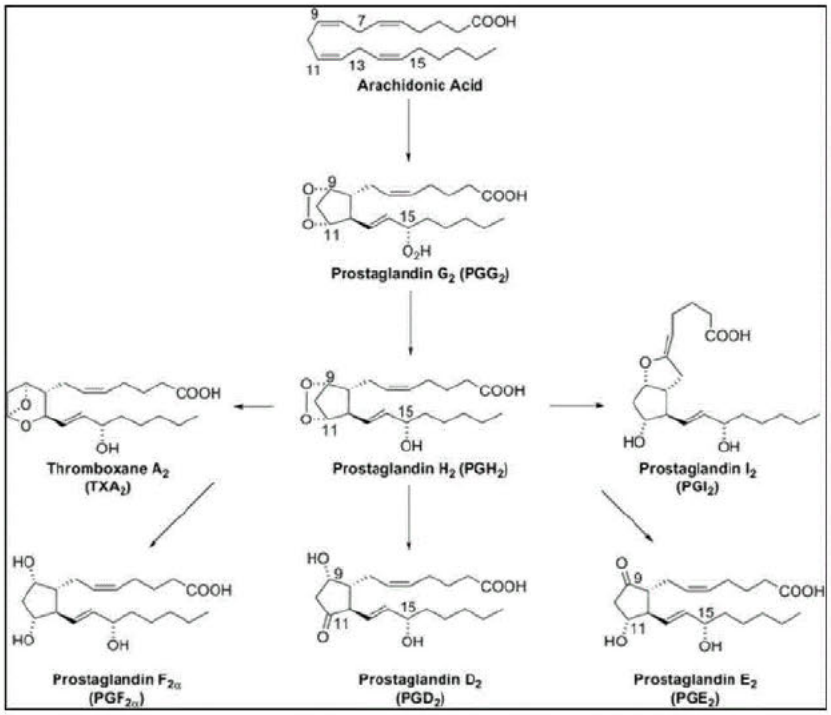


Figure #2: Complete arachidonic acid metabolism into prostaglandins (including PGE_2)

b. Research Question

Acetylsalicylic acid ($C_9H_8O_4$), commonly known as aspirin, is one of the most widely known NSAIDs in medicine, used for blood thinning and other anti-inflammatory purposes. This is due to its inhibition of prostaglandin formation, where prostaglandins are sensed by prostanoid receptors to induce inflammation (Ricciotti and FitzGerald). Experimental data consistently shows that aspirin inhibits the activity of cyclooxygenase-1 at relatively low concentrations (Golstov et al.). Due to this known property of aspirin, its inhibition is often used as a baseline to compare it to drugs with unknown inhibitory activity. One of such drugs is acetaminophen ($C_8H_9NO_2$). Acetaminophen is a fever reducing drug like aspirin. However, it is not classified as NSAID despite its analgesic properties (Hinz et al.). Therefore, it will be

important to investigate if acetaminophen has inhibitory effects on COX-1 similar to aspirin, to determine why acetaminophen is not classified as an NSAID. In other words: **Is acetaminophen as effective as aspirin (acetylsalicylic acid) as an inhibitor of cyclooxygenase-1 (COX-1) catalysis of arachidonic acid metabolism, determined by calculating the IC_{50} values?**

The common metric for identifying the effectiveness of an inhibitor on a reaction is by calculating its inhibition constant (IC_{50}), or the concentration that inhibits the reaction by 50%. To measure how much each concentration of each drug has inhibited COX-1 activity, each rate of reaction will be compared to the maximum rate, the rate of arachidonic acid metabolism without inhibitors. Different concentrations of aspirin and acetaminophen will be added to the reaction to plot the relation between inhibitor concentration and the rate as a percentage of the maximum inhibition.

2. Methodology

a. Experimental Approach

The experimental work was conducted at the American University of Beirut's Department of Biochemistry and Molecular Genetics. COX-1 used in each reaction was obtained from human embryonic kidney 293 cells (HEK-293), which is cultured in 12-well plates at around 200,000 cells per well prior to the experiment. The arachidonic acid and inhibitor solutions will be diluted in Hanks' Buffer, a salt buffer solution that will regulate the pH of each medium at 7.4 (El-Achkar et al.). At the end of each experiment, the media from each of the wells will be collected and frozen for analysis.

Before experimenting with inhibitors, preliminary experiments will be performed. The first will involve a constant concentration of arachidonic acid reacting in the presence of COX-1 in the 12-well plate, with each sample removed at different times. This is to determine the time around which the reaction is completed, which was found to be 30 minutes. The second experiment involved varying concentrations of arachidonic acid left to react for 30 minutes to find the concentration of arachidonic acid that saturates the reaction (where increasing the concentration will not increase the rate of reaction). The products will be collected from each well, stopping the reaction by removing the substrate from the enzyme, and frozen for analysis. The saturated concentration of arachidonic acid during metabolism was found to be 20 $\mu mol.dm^{-3}$, which was used for the reactions inhibition by aspirin and acetaminophen. [1]

[1] See Appendices C and D

For the primary experiment, solutions of varying concentrations of the drugs will be made with serial dilutions in Hanks' Buffer and the cells will be treated in these solutions 30 minutes prior to the experiment. Serial dilutions are important to maintain the same proportion of aspirin and acetaminophen in each of the solutions and to avoid random error when making dilutions. As the primary solution of each drug was diluted in ethanol, it is important to make intermediary solutions of each drug to regulate the ethanol content of the subsequent solutions, such that it does not affect the activity of the reaction. The intermediary solutions will be stored in 2.0 cm^3 cryovials to prevent ethanol evaporation. Each reaction will be left for 30 minutes reacting with 20 $\mu mol.dm^{-3}$ of arachidonic acid. To avoid pipetting small quantities of arachidonic acid, an intermediary solution will also be formed prior to adding to the HEK-293 cells and the inhibitor solutions.

As this reaction takes place with little to no changes to the reactants' or products' macroscopic properties, spectrophotometric analysis in the form of an enzyme immunoassay (EIA) is performed in a 96-well microplate to quantify concentrations of PGE_2 in each sample to calculate the rates of reaction. This procedure is regularly used at the AUB laboratory, with some modifications to fit the specifics of this experiment. The EIA measures the unknown concentration of PGE_2 that binds to anti-PGE_2 antibodies, as PGE_2 competes with a PGE_2-tracer linked to the enzyme acetylcholinesterase. After incubation, Ellman's Reagent, the substrate of acetylcholinesterase, is added to each of the wells and reacts to form a yellow product. This yellow color is directly related to the concentration of PGE_2 tracer and inversely related to the concentration of PGE_2 (Pradelles et al. "Enzyme Immunoassays: An Alternative to Radioimmunoassay").

The method used to quantify the PGE_2 concentrations is also outlined in the paper that published the EIA protocol. To calculate the unknown PGE_2 concentrations obtained from the reactions, they will be compared to solutions of known PGE_2 concentrations, starting with 250 $pg.cm^{-3}$ and serially diluted by half to reach 1.95 $pg.cm^{-3}$, through a standard curve. Based on preliminary experiments conducted, the PGE_2 samples obtained from the experiments will exceed this range, therefore each of the samples must be diluted by a certain extent before conducting an analysis EIA.

The first row of the microplate will also be used to correct and calculate concentrations, by preparing Blank, Non-specific Binding, and B_0 values. The Blank value is the smallest possible absorbance value. In two wells of the microplate, 150 mm^3 of colorless EIA Buffer solution is added. These wells should yield the lowest absorbance values, and by subtracting it from every subsequent absorbance value each data point can be corrected. The Non-specific Binding (NSB) value is the absorbance of PGE_2 tracer non-specifically binding to the walls of the microplate and not the antibody. In two wells 100 mm^3 of EIA Buffer and 50 cm^3 of the PGE_2 tracer is added. Trace amounts of PGE_2 tracer bind to the walls of the microplate instead of the anti-PGE_2 antibody in all 96 wells. The NSB value is subtracted from the B_0 value to account for this non-specific binding. B_0 is the maximum absorbance value. 50 mm^3 each of EIA Buffer, PGE_2 tracer, and anti-PGE_2 antibody will be added in four wells of the microplate. This yields the strongest color possible, as all the tracer binds to the antibody, leaving none for the PGE_2. The B_0 average is crucial to the success of the EIA; subsequent absorbance values are calculated as percentages of the B_0 values. This is why the combination of solutions for B_0 values is placed in four separate wells instead of two, in the event of outliers caused by random error. All solutions will be placed under a plate reader,

which will measure the absorbance of yellow light at 414 nm, the wavelength of its complementary color, violet (Pradelles et al. "Enzyme Immunoassays of Eicosanoids Using Acetylcholinesterase").

b. Variables

Table #1: Independent Variables

Variable	Method of manipulation
Concentration of inhibitors (acetaminophen and aspirin)	Solutions of acetaminophen and aspirin will be serially diluted to give each a different concentration, which will be used to explore the inhibition of COX-1.

Table #2: Dependent Variables

Variable	Method of Measurement
PGE_2 concentration	The EIA will be performed to calculate the concentration product formed by each reaction.
Inhibition	Each rate of reaction will be compared to the maximum rate to calculate inhibition.

Table #3: Controlled Variables

Variable	Effect of the variable if left uncontrolled	Method of control
Reaction duration	Different reaction duration will vary the concentration of PGE_2 collected for the same concentration of arachidonic acid.	The duration for all reactions was set at 30 minutes, based on preliminary trials conducted.
Concentration of arachidonic acid	Different concentrations of arachidonic acid will lead to different rates of reaction for the same concentration of inhibitor ("10.2: The Equations of Enzyme Kinetics").	The concentration of arachidonic acid was set at 20 $\mu mol.dm^{-3}$. The reaction is saturated at this concentration, according to a preliminary experiment.
Concentration of COX-1	Different concentrations of COX-1 will lead to different rates of reactions for experiments of the same concentration of arachidonic acid and inhibitor ("10.2: The Equations of Enzyme Kinetics").	Around 200,000 HEK-293 cells were cultured per reaction, to regulate the concentration of COX-1 among the reactions.
Acidity level	Different pHs will affect the activity of the enzyme: a pH other than the optimal pH of the reaction will not allow the enzyme to work at maximum capacity ("1.18 Enzyme Function").	The arachidonic acid and inhibitors were diluted in Hanks' Buffer, a salt buffer solution that maintains the pH at 7.4.
Temperature of the system	Different temperatures will affect the activity of COX-1: a high temperature will denature the enzyme, whereas a low temperature will render the enzyme inactive ("1.18 Enzyme Function").	All reactions took place in a cell culture incubator, at a constant temperature of 37°C.

c. Primary Experiment Method

The detailed protocol for this experiment was developed with the help of the AUB Biochemistry Department, who gave guidance on the appropriate choice of methods and apparatuses. This experiment method is typically used when investigating the effect of drugs on the activity of enzymes secreted by cells.

1. Culture HEK-293 cells in four 12-well plates at 200,000 cells per well.
2. Prepare and label Eppendorf microtubes to collect samples.
3. Suspend 15.4 mg of acetaminophen to 679 mm^3 of ethanol to form a 0.150 $mol.dm^{-3}$ solution.
4. Add 110.0 mm^3 of 0.150 $mol.dm^{-3}$ acetaminophen into 390 mm^3 of ethanol to form a 0.0330 $mol.dm^{-3}$ intermediary solution of acetaminophen.
5. Add 10.00 mm^3 of 0.200 $mol.dm^{-3}$ aspirin into 90.0 mm^3 of ethanol to form a 20.0 $mmol.dm^{-3}$ intermediary solution of aspirin.
6. Prepare 75.0 cm^3 of Hanks' Buffer.[2]
7. Add 15.00 mm^3 of 20.0 $mmol.dm^{-3}$ aspirin into 10.00 cm^3 of Hanks' Buffer to form a 30.0 $\mu mol.dm^3$ aspirin solution.
8. Serially dilute the 30.0 $\mu mol.dm^3$ aspirin solution in Hanks' Buffer to form aspirin solutions of concentration 10.0, 3.00, 1.00, 0.300, and 0.100 $\mu mol.dm^3$.
9. Add 15.00 mm^3 of 33.0 $mmol.dm^{-3}$ acetaminophen into 10.00 cm^3 of Hanks' Buffer to form a 50.0 $\mu mol.dm^3$ aspirin solution.
10. Serially dilute the 50.0 $\mu mol.dm^3$ acetaminophen solution in Hanks' Buffer to form acetaminophen solutions of concentrations 20.0, 10.0, 5.00, 2.00, 1.00 $\mu mol.dm^3$.[3]

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[2] See Appendix B.

[3] For exact preparation of the solutions found in step 8 and 10, see Appendix B.

11. Pipette 1 cm^3 of each of the aspirin and acetaminophen solutions into 3 wells each.
12. Incubate samples at 37°C for 30 minutes.
13. Add 24.0 mm^3 of 50.0 $mmol.dm^{-3}$ arachidonic acid to form a 0.800 $mmol.dm^{-3}$ intermediary arachidonic acid solution.
14. Remove samples from the incubator.
15. Pipette 1 cm^3 of 0.15% ethanol into 6 wells each (3 for negative control, 3 for positive control).
16. Pipette 25.0 mm^3 of 0.800 $mmol.dm^{-3}$ arachidonic acid in each well except the negative control wells (this will result in 20.0 $\mu mol.dm^3$ arachidonic acid in each well).
17. Incubate at 37°C for 30 minutes.
18. Remove samples from the incubator and collect in Eppendorf microtubes.
19. Freeze samples at -20°C for future enzyme immunoassay.

To perform an enzyme immunoassay:

1. Prepare the 96-well microplate (pre-made in the lab)
2. Dilute a PGE_2 sample of known concentration (250 $pg.cm^{-3}$) by adding 500 mm^3 of it into 500 mm^3 of EIA Buffer, and continue halving the concentration of PGE_2 concentration until the 8th dilution.
3. Dilute the samples of unknown PGE_2 concentration (the degree to which each sample is to be diluted is found in tables 10.1 through 10.3).
4. Pipette 150.0 mm^3 of EIA Buffer into 2 wells each for Blank.
5. Pipette 100.0 mm^3 of EIA Buffer and 50.0 of PGE_2 tracer into 2 wells each for NSB.
6. Pipette 50.0 mm^3 of EIA Buffer, PGE_2 tracer, and anti-PGE_2 antibody into 4 wells each for B_0.

7. Pipette 50.0 mm^3 of each of the known concentrations of PGE_2 (2 wells for each concentration).
8. Pipette 50.0 mm^3 of each PGE_2 sample into each well.
9. Multipipette 50.0 cm^3 anti-PGE_2 antibody and PGE_2 tracer in each well.
10. Incubate plate at 4°C for 24 hours.
11. Empty the plate and wash each well with 300 mm^3 of Wash Buffer (repeat 3 times)
12. Multipipette 200 mm^3 of Ellman's reagent into each well.
13. Incubate the plate in darkness at room temperature for 4 hours.
14. Analyze the plate in the plate reader at 414 nm.

3. Raw Data (Spectrophotometric)

Table #4.1: Negative control absorbance values (0 µmol.dm^{-3})

	Absorbance (± 0.001 mAU)		
	Trial 1	Trial 2	Trial 3
Negative Control	0.156	0.251	0.246

Table #4.2: Variation of absorbance as a function of aspirin concentration

[Aspirin] (µmol.dm^{-3})	Absorbance (± 0.001 mAU)		
	Trial 1	Trial 2	Trial 3
0.00	0.146	0.147	0.143
0.100	0.143	0.147	0.237
0.300	0.175	0.346	0.166
1.00	0.325	0.158	0.150
3.00	0.150	0.160	0.146
10.0	0.229	0.185	0.218
30.0	0.130	0.104	0.164

Table #4.3: Variation of absorbance as a function of acetaminophen concentration

[Acetaminophen] (µmol.dm^{-3})	Absorbance (± 0.001 mAU)		
	Trial 1	Trial 2	Trial 3
0.00	0.146	0.147	0.143
1.00	0.158	0.132	0.152
2.00	0.147	0.150	0.133
5.00	0.116	0.151	0.153
10.0	0.441	0.158	0.144
20.0	0.151	0.150	0.158
50.0	0.150	0.150	0.147

Table #5.1: Variation of absorbance as a function of PGE_2 concentration (for standard curve)

$[PGE_2]$ ($pg.cm^{-1}$)	Absorbance (± 0.001 mAU)	
	Value 1	Value 2
250	0.132	0.125
125	0.137	0.140
62.5	0.165	0.150
31.3	0.199	0.199
15.6	0.216	0.216
7.81	0.323	0.314
3.91	0.379	0.357
1.95	0.428	0.419

Table #5.2: Variation of absorbance as a function of qualification (for standard curve)

Qualification	Absorbance (± 0.001 mAU)			
	Value 1	Value 2	Value 3	Value 4
Blank	0.104	0.101	-	-
NSB	0.097	0.097	-	-
B_0	0.425	0.429	0.432	0.443

4. Analysis

a. Absorbance Corrections

To calculate the concentrations of the samples, the absorbance-concentration graph for known concentrations of PGE_2 is constructed. All absorbance values are first corrected by subtracting the average Blank value, as it is the smallest absorbance value possible that should be obtained. This must take place before averaging the absorbance values from all three trials.

$$Average\ Blank = \frac{0.104+0.101}{2} = 0.103 \pm 0.001\ mAU$$

For Aspirin and Acetaminophen 0 µmol.dm^{-3} (1): $0.146 - 0.103 = 0.043$

Table #6.1: Negative control (corrected) absorbance values (0 µmol.dm^{-3})

	Absorbance (± 0.002 mAU)			Average Absorbance
	Trial 1	Trial 2	Trial 3	
Negative Control	0.053[4]	0.148	0.143	0.146

Table #6.2: Variation of (corrected) absorbance as a function of aspirin concentration

[Aspirin] (µmol.dm^{-3})	Absorbance (± 0.002 mAU)			Average Absorbance
	Trial 1	Trial 2	Trial 3	
0	0.043	0.041	0.040	0.041
0.100	0.040	0.044	0.134	0.042
0.300	0.072	0.243	0.063	0.068
1.00	0.222	0.055	0.047	0.051
3.00	0.047	0.057	0.043	0.049
10.0	0.126	0.082	0.115	0.108
30.0	0.027	0.001	0.061	0.044

[4] All the values colored in red have been identified as outliers from the trend, and thus have been removed from average calculations.

Table #6.3: Variation of (corrected) absorbance as a function of acetaminophen concentration

[Acetaminophen] ($\mu mol.dm^{-3}$)	Absorbance (± 0.002 mAU)			Average Absorbance
	Trial 1	Trial 2	Trial 3	
0	0.043	0.041	0.040	0.041
1.00	0.055	0.029	0.049	0.052
2.00	0.044	0.047	0.030	0.046
5.00	0.013	0.048	0.050	0.049
10.0	0.338	0.055	0.041	0.048
20.0	0.048	0.047	0.055	0.050
50.0	0.047	0.047	0.044	0.046

Table #7.1: Variation of (corrected) absorbance as a function of PGE_2 concentration (for standard curve)

$[PGE_2]$ ($pg.cm^{-1}$)	Absorbance (± 0.002 mAU)		Average Absorbance
	Value 1	Value 2	
250	0.029	0.022	0.026
125	0.034	0.037	0.036
62.5	0.062	0.047	0.055
31.3	0.096	0.096	0.096
15.6	0.113	0.113	0.113
7.81	0.220	0.211	0.216
3.91	0.276	0.254	0.265
1.95	0.325	0.316	0.321

Table #7.2: Variation of (corrected) absorbance as a function of qualification (for standard curve)

Qualification	Absorbance (± 0.001 mAU)			
	Value 1	Value 2	Value 3	Value 4
NSB	-0.006	-0.006	-	-
B_0	0.322	0.326	0.329	0.340

b. Standard Curve Construction

According to Pradelles' method, these corrected absorbance values are divided by the average B_0 value to construct the standard curve. All values (B) and the B_0 are further corrected by subtracting the average NSB.

$$Average\ NSB = \frac{-0.006-0.006}{2} = -0.006 \pm 0.001$$

$$Average\ B_0 = \frac{0.322+0.326+0.329+0.340}{4} = 0.329 \pm 0.009$$

$$B/B_0 = \frac{B-NSB}{B_0-NSB}$$

$$For\ Aspirin\ and\ Acetaminophen\ 0\ \mu mol.dm^{-3}: \frac{0.041+0.006}{0.329+0.006} = 0.140$$

To linearize the standard curve for more convenient concentration calculation, the average logit of B/B_0 plotted against concentration in log scale.

$$Logit = log\left(\frac{B/B_0}{1-B/B_0}\right)$$

$$Sample\ calculation\ for\ PGE_2\ 250\ pg.cm^{-1}: log\left(\frac{0.096}{1-0.096}\right) = -0.976$$

Table #8.1: Negative control B/B_0 values (0 $\mu mol.dm^{-3}$)

	Average Absorbance	Average B/B_0	Average Logit (B/B_0)
Negative Control	0.146	0.454	-0.081

Table #8.2: Variation of B/B_0 as a function of aspirin concentration

[Aspirin] ($\mu mol.dm^{-3}$)	Average Absorbance	Average B/B_0	Average Logit (B/B_0)
0	0.041	0.140	-0.787
0.100	0.042	0.143	-0.777
0.300	0.068	0.221	-0.547
1.00	0.051	0.170	-0.688
3.00	0.049	0.164	-0.707
10.0	0.108	0.340	-0.287
30.0	0.044	0.149	-0.756

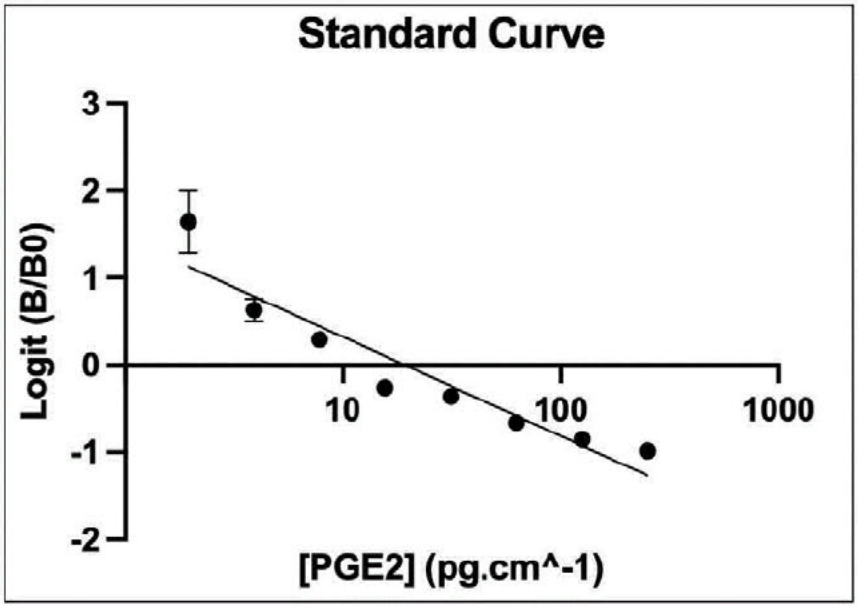
Table #8.3: Variation of B/B_0 as a function of acetaminophen concentration

[Acetaminophen] ($\mu mol.dm^{-3}$)	Average Absorbance	Average B/B_0	Average Logit (B/B_0)
0	0.041	0.140	-0.787
1.00	0.052	0.173	-0.679
2.00	0.046	0.155	-0.736
5.00	0.049	0.164	-0.707
10.0	0.048	0.161	-0.716
20.0	0.050	0.167	-0.697
50.0	0.046	0.155	-0.736

Table #9.1: Variation of B/B_0 as a function of PGE_2 concentration (for standard curve)

[PGE2] ($pg.cm^{-3}$)	Average Absorbance	Average B/B_0	Average Logit (B/B_0)
250	0.026	0.096	-0.976
125	0.036	0.125	-0.844
62.5	0.055	0.182	-0.652
31.3	0.096	0.304	-0.359
15.6	0.113	0.355	-0.259
7.81	0.216	0.663	0.293
3.91	0.265	0.809	0.627
1.95	0.321	0.976	1.611

Graph #1: Variation of logit B/B_0 as a function of log PGE_2 concentration



Curve Equation: $y = -1.14x + 1.458$

This is a curve in the form y = mx + b, where y is the logit of B/B_0 and x is PGE_2 concentration. It is now possible to calculate the unknown concentrations of all the samples, simply by isolating x, raising it to a power of 10 since a logarithmic scale is taken for the x-axis, and multiplying all EIA concentrations by the dilution factor of each sample.

c. Concentration Calculations

$$Concentration = 10^{\frac{Logit\ (B/B0) - 1.458}{-1.14}} * Dilution\ Factor$$

For Aspirin and Acetaminophen 0 µmol.dm^{-3} (1):

$$C = 1000 * 10^{\frac{-0.787 - 1.458}{-1.14}} = 93.2\ ng.cm^{-3}$$

Table #10.1: Negative control concentration values (0 µmol.dm^{-3})

	Average Logit	EIA [PGE] (ng.cm^{-3})	Dilution Factor	[PGE] (ng.cm^{-3})
Negative Control	-0.102	0.0234	10	0.234

Table #:10.2 Variation of PGE_2 concentration as a function of aspirin concentration

[Aspirin] (µmol.dm^{-3})	Average Logit	EIA [PGE] (ng.cm^{-3})	Dilution Factor	[PGE] (ng.cm^{-3})
0	-0.787	0.0932	1000	93.2
0.100	-0.777	0.0912	1000	91.2
0.300	-0.547	0.0574	1000	57.4
1.00	-0.688	0.0763	1000	76.3
3.00	-0.707	0.0792	1000	79.2
10.0	-0.287	0.0340	300	10.2
30.0	-0.756	0.0875	200	17.5

Table #10.3: Variation of PGE_2 concentration as a function of acetaminophen concentration

[Acetaminophen] ($\mu mol.dm^{-3}$)	Average Logit	EIA [PGE] ($ng.cm^{-3}$)	Dilution Factor	[PGE] ($ng.cm^{-3}$)
0	-0.787	0.093	1000	93.2
1.00	-0.679	0.075	1000	74.9
2.00	-0.736	0.084	1000	84.0
5.00	-0.707	0.079	1000	79.2
10.0	-0.716	0.081	1000	80.8
20.0	-0.697	0.078	1000	77.8
50.0	-0.736	0.084	1000	84.0

d. Rate and Inhibition Calculations

The rate for each reaction is found by dividing the change in concentration of PGE_2 over the reaction time (30 mins). The change in concentration of PGE_2 in this case is the concentration of PGE_2 obtained, assuming that the initial concentration is always 0.

$$Rate = \frac{Change\ in\ concentration}{Time}$$

Each rate of reaction is now compared as a percent of the average maximum rate of reaction for arachidonic acid metabolism (rate for 0 $\mu mol.dm^{-3}$ aspirin and acetaminophen) to find how much each drug has inhibited the reaction.

$$Maximum\ Rate = \frac{93.2}{30} = 3.11\ ng.cm^{-3}.min^{-1}$$

$$Inhibition: Percentage\ of\ maximum\ rate = \frac{Rate}{3.11} * 100$$

$$For\ Aspirin\ 0.100\ \mu mol.dm^{-3}: Percentage\ of\ maximum\ rate = \frac{3.04}{3.11} * 100 = 98.0\ \%$$

Table #11.1: Variation of PGE_2 rate of formation as a function of aspirin concentration

[Aspirin] ($\mu mol.dm^{-3}$)	Rate ($ng.cm^{-3}.min^{-1}$)	% of Maximum Rate
0	3.11	100
0.100	3.04	98.0
0.300	1.91	61.6
1.00	2.54	81.9
3.00	2.64	85.1
10.0	0.340	10.9
30.0	0.583	18.8

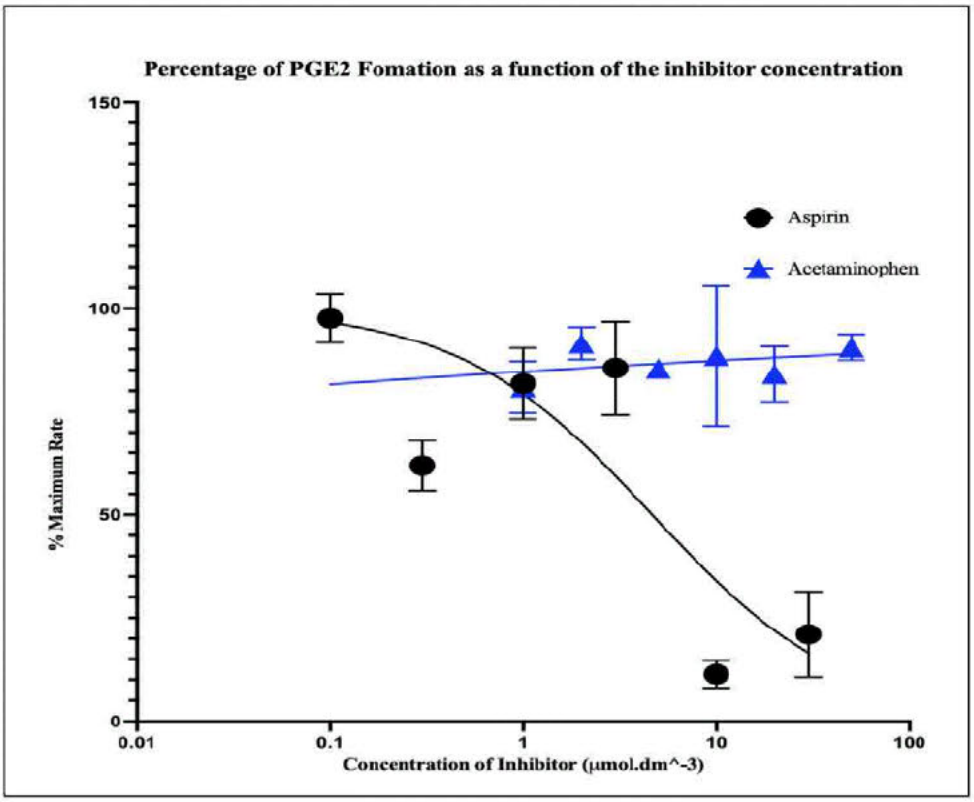
Table #11.2: Variation of PGE_2 rate of formation as a function of acetaminophen concentration

[Acetaminophen] ($\mu mol.dm^{-3}$)	Rate ($ng.cm^{-3}.min^{-1}$)	% of Maximum Rate
0	3.11	100
1.00	2.50	80.4
2.00	2.80	90.1
5.00	2.64	85.0
10.0	2.69	86.7
20.0	2.59	83.5
50.0	2.80	90.1

The concentration of inhibitors is plotted against the average inhibition with a sigmoidal function fitting using PRISM software to find the IC_{50} of each drug. The sigmoidal function includes the IC_{50} of each drug as coefficient in the equation, and includes the Hill Slope which indicates how steep the curve is. The equation for this is therefore (where y is the percentage of the maximum rate and x is the concentration of each drug):

$$y = \frac{100}{1 + (IC_{50}/x)^{Hillslope}}$$

Graph #2: Variation of percent-PGE_2 formation as a function of aspirin and acetaminophen concentration.



$$For\ Aspirin: y = \frac{100}{1 + (4.65/x)^{-0.869}};\ R^2 = 0.7313$$

For Acetaminophen: No sigmoidal equation confidently determined

The IC_{50} value for each drug, as calculated by PRISM software, are presented in table #12.

Table #12: IC_{50} values

Drug	IC_{50} ($\mu mol.dm^{-3}$)
Aspirin	4.65
Acetaminophen	Undetermined

Since the rate of PGE_2 formation in the presence of acetaminophen did not decrease lower than 80%, it is impossible to calculate an accurate IC_{50} value.

5. Conclusion

As previously mentioned, comparing the effectiveness of the inhibitors would be done by comparing the IC_{50} values of each drug. As exhibited by Graph #2 and Table #12, aspirin's IC_{50} is 4.65 $\mu mol.dm^{-3}$, while no IC_{50} can be calculated for acetaminophen because it did not even reach 50% inhibition of COX-1.

As expected, the graph shows the rate of arachidonic acid metabolism significantly decreased as the concentration for aspirin increased, while the arachidonic acid metabolism rate slightly decreased as the acetaminophen concentration increased. While it appears that both drugs affected the reaction rate, only the aspirin-treated reactions exhibited inhibition above 50%. Therefore, the experiment shows that aspirin is a more effective COX-1 inhibitor than acetaminophen, when comparing IC_{50} values. Furthermore, this experiment supports the premise that acetaminophen is not an NSAID, since it could not reliably inhibit the activity of COX-1.

6. Evaluation of Results

a. Propagation of Uncertainties

$$\%Uncertainty = Previous\ Uncertainty + ((\frac{Uncertainty\ of\ Pipette}{Volume\ of\ solution}) + (\frac{Uncertainty\ of\ Pipette}{Volume\ of\ Hanks'\ Buffer})) * 100$$

Note that previous uncertainty is the uncertainty of the solution that was prepared immediately before the subsequent solution.

$$For\ Aspirin\ 30.0\ \mu mol.dm^{-3}:\ 0.21\% + (\frac{0.01}{15.00}) * 100 + (\frac{0.05}{10.00}) * 100 = 5\%$$

Tables #13.1a: Uncertainty propagation of aspirin solutions prepared

[Aspirin] (mmol.dm^{-3})	Percent Uncertainty
20.0	0.21

Tables #13.1b: Uncertainty propagation of aspirin solutions prepared

[Aspirin] (µmol.dm^{-3})	Percent Uncertainty
0.100	19
0.300	15
1.00	12
3.00	8
10.0	5
30.0	0.78

Table #13.2a: Uncertainty propagation of acetaminophen solutions prepared

[Aspirin] (mol.dm^{-3})	Percent Uncertainty
0.150	0.80
0.0330	1.14

Table #13.2b: Uncertainty propagation of acetaminophen solutions prepared

[Acetaminophen] (μmol.dm^{-3})	Percent Uncertainty
1.00	13
2.00	11
5.00	8
10.0	6
20.0	4
50.0	1.7

$$\%Uncertainty = (2 * (\frac{0.003}{B-NSB} + \frac{0.002}{B_0-NSB}) + \frac{Uncertainty\ of\ time}{Time} + \frac{Uncertainty\ of\ maximum\ rate}{Maximum\ Rate}) * 100$$

$$Uncertainty\ of\ Maximum\ Rate = (2 * (\frac{0.003}{B-NSB} + \frac{0.002}{B_0-NSB}) + \frac{Uncertainty\ of\ time}{Time}) * Maximum\ Rate$$

$$Uncertainty\ of\ Maximum\ Rate = (2 * (\frac{0.003}{0.047} + \frac{0.002}{0.335}) + \frac{1/60}{30}) * 3.11 = \pm 0.43\ ng.cm^{-1}.min^{-1}$$

$$For\ Aspirin\ 30\ \mu mol.dm^{-3} = (2 * (\frac{0.003}{0.098} + \frac{0.002}{0.335}) + \frac{1/60}{30} + \frac{0.43}{3.11}) * 100 = 27\%$$

Table #14.1: Uncertainty propagation of aspirin inhibition

[Aspirin] (μmol.dm^{-3})	Inhibition Percent Uncertainty
0.100	29.4
0.300	23.9
1.00	26.8
3.00	27.3
10.0	20.6
30.0	28.7

Table #14.2: Uncertainty propagation of acetaminophen inhibition

[Acetaminophen] ($\mu mol.dm^{-3}$)	Inhibition Percent Uncertainty
1.00	27
2.00	28
5.00	27
10.0	28
20.0	27
50.0	28

b. Discussion

While this experiment indicates that aspirin is a better COX-1 inhibitor than acetaminophen, the medium in which this exploration took place should be taken into consideration. The HEK-293 cells used in this experiment naturally produce PGE_2 without the addition of arachidonic acid (Tebbenkamp et al.). This partly explains why the negative control performed alongside the experiment measured some concentrations of PGE_2 without any arachidonic acid. These values were small, so this suggests that the trace amounts of PGE_2 in the negative control could simply be the result of a random error.

HEK-293 cells overexpress enzymes other than COX-1, some of which could be inhibited by aspirin or acetaminophen, so it is unclear in this case whether the conclusion of this experiment is true (Tebbenkamp et al.). A more appropriate experimental approach that was initially envisioned for this exploration would be to isolate the COX-1 enzyme to better investigate the effect of aspirin and acetaminophen on COX-1, as it would make for a more specific experiment. Unfortunately, due to material shortages in Beirut, there was little feasibility in this plan, so this current method devised was the best one given the situation.

The enzyme immunoassay technique, while providing precise results, is hypersensitive, meaning a small change in the absorbance of the sample could vary the concentration analyzed massively. As I have little experience in laboratory work which means a greater tendency to make mistakes, those mistakes played a factor in the EIA's concentration readings, affecting the final inhibition calculations. In an ideal setting, the EIA for the primary experiment would have been repeated to calculate the inter-plate error and more usable data points. However, there is a high monetary cost associated with the reagents needed to perform one EIA, and seeing as this experiment was conducted in a country undergoing a financial crisis, this also was not feasible.

During the analysis of the results, some inhibition values for aspirin and acetaminophen had to be excluded as they were clear outliers for the rest of the results. In addition, the points on both the aspirin and acetaminophen curves included some anomalies that were far from the trendline. In acetaminophen's case, the concentration vs. inhibition graph shows an upward trend, suggesting that the rate of formation increases with an increase in acetaminophen concentration, perhaps solidifying that acetaminophen provides little COX-1 inhibition. In the case of aspirin, while the results did form a recognizable trendline, the coefficient of correlation (R^2) was 0.7313, indicating a very weak correlation between the x and y values. However, a previous study calculated the IC_{50} of aspirin inhibition as 2 $\mu mol.dm^{-3}$, and my IC_{50} value is comparable to those established by other enzymologists, suggesting that my data set is somewhat accurate (Gorkov et al.).

While it may be concluded from the results of this experiment that acetaminophen has

no anti-inflammatory properties, this experiment doesn't present the full picture. As mentioned previously, NSAIDs inhibit one of the oxygenase enzymes, one of which is COX-2, another isoform of cyclooxygenase in humans. Acetaminophen at high concentrations has been found to significantly inhibit COX-2 activity when tested on samples of human plasma. In addition, a previous study did find that acetaminophen could inhibit COX-1 at high concentrations, with an IC_{50} of 113.7 $\mu mol.dm^{-3}$ (Hinz et al.). Therefore, the question of whether acetaminophen is NSAID can be further investigated.

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8. Appendices

Appendix A: Materials List

Reagents:

- Arachidonic Acid (10.0 and 50.0 mmol.dm^{-3})
- HEK-293 cells overexpressing human COX-1
- Hanks' Buffer (5x Concentration)
- Acetaminophen
- Acetylsalicylic acid (0.200 mol.dm^{-3})
- Ethanol
- Sodium bicarbonate (0.400 mol.dm^{-3})
- Bovine Serum Albumin
- Dilute hydrochloric acid and sodium hydroxide solutions
- Distilled water
- EIA Buffer
- Anti Prostaglandin E_2 antibody (1:30000 concentration)
- Prostaglandin E_2 tracer
- EIA Wash Buffer
- Ellman's reagent (1:30 concentration)

Instruments:

- 200-1000 mm^3 pipette w/tips (± 1 mm^3)
- 20-200 mm^3 pipette w/tips (± 0.1 mm^3)
- 2-20 mm^3 pipette w/tips (± 0.01 mm^3)
- 1-10 mm^3 pipette w/tips (± 0.01 mm^3)
- Eppendorf repeater pipette w/ 50 - 300 mm^3 settings
- 10 cm^3 disposable pipettes w/pipette aid (± 0.05 cm^3)
- 5 cm^3 disposable pipettes w/pipette aid (± 0.05 cm^3)
- 12-well plates
- 96 well microplates

- Cell culture incubator (37°C and 5% CO_2)
- Cell culture hood
- Aspirator
- 1.5 cm^3 Eppendorf microtubes
- 2 cm^3 cryovials
- 50 cm^3 test tubes
- 15 cm^3 test tubes
- 5 cm^3 test tubes
- 100 cm^3 beakers
- 400 cm^3 beakers
- 100.0 cm^3 graduated cylinder (± 0.5 cm^3)
- Magnetic stirrer
- Hot plate
- Mettler Balance (± 0.1 mg)
- Weighing paper
- Spatula
- pH-meter (± 0.01 pH) w/calibrator solutions
- Plate reader (± 0.001 mAU)

Appendix B: Supplements to Experimental Method

To make 75 cm^3 of Hanks' Buffer:

1. Thaw 15.0 cm^3 of 5x concentrated Hanks' Buffer.
2. Add 75.0 mm^3 of sodium bicarbonate solution to the concentrated Hanks' Buffer in a beaker and add some distilled water.
3. Dissolve 75.0 mg of Bovine Serum Albumin on the surface of the Hanks' Buffer solution.
4. Calibrate the pH-meter by placing its probe in solutions of known pH 7 and 10.
5. Measure the acidity of the Hanks Buffer (the desired pH is 7.4).
    a. If the pH is too high, gradually add dilute hydrochloric acid.
    b. If the pH is too low, gradually add dilute sodium hydroxide.
6. Once the desired pH is reached, add distilled water until the total volume of the solution is 75 cm^3.

Table #15.1: Specifics of serially diluted aspirin solutions (as found in step 8 of Section 2C)

Volume of previous solution (cm^3)	Volume of Hanks' Buffer (cm^3)	Concentration of new solution ($\mu mol.dm^3$)	Volume of new solution (cm^3)
2.00	4.00	10.0	6.00
2.00	4.70	3.00	6.70
2.00	4.00	1.00	6.00
2.00	4.70	0.300	6.70
2.00	4.00	0.100	6.00

Table #15.2: Specifics of serially diluted acetaminophen solutions (as found in step 10 of Section 2C)

Volume of previous solution (cm^3)	Volume of Hanks' Buffer (cm^3)	Concentration of new solution ($\mu mol.dm^3$)	Volume of new solution (cm^3)
4.00	6.00	20.0	10.00
4.00	4.00	10.0	8.00
4.00	6.00	5.00	10.00
4.00	4.00	2.00	8.00
4.00	6.00	1.00	10.00

Appendix C: Preliminary Experiment #1

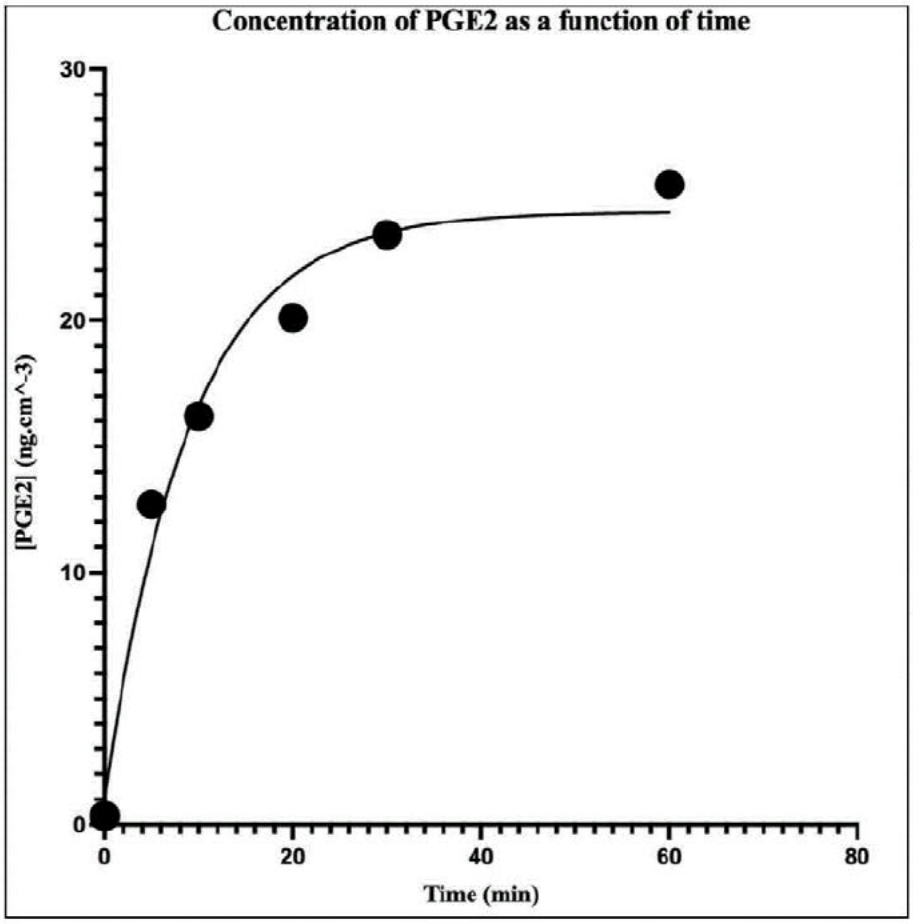
Method:

1. Culture HEK-293 cells in four 12-well plates at 200,000 cells per well.
2. Prepare and label Eppendorf microtubes to collect samples.
3. Prepare 50 cm^3 of Hanks's Buffer solution
4. Add 25.0 mm^3 of 10.0 $mmol.dm^{-3}$ arachidonic acid into 25.0 cm^3 of Hanks Buffer in a test tube to form a 10.0 $\mu mol.dm^{-3}$ arachidonic acid solution
5. Pipette 1 cm^3 of Hanks Buffer per well into the first 3 wells of a 12-well plate for negative control
6. Pipette 1 cm^3 of 10.0 $\mu mol.dm^{-3}$ arachidonic acid in each well of the 12-well plate (excluding the first 3).
7. Incubate at 37°C in the cell culture incubator.
8. After 5 minutes, remove the plates from the incubator and collect samples from the second column of three wells into Eppendorf microtubes and put the plates back in the incubator.
9. Repeat step 8 at 10, 20, 30, 45, and 60 minutes.
10. Freeze samples at -20°C for future EIA.

Table #16: Variation of PGE_2 formation as a function of time

Time (min)	$[PGE_2]$ ($ng.cm^{-3}$)			Average $[PGE_2]$ ($ng.cm^{-3}$)
	Trial 1	Trial 2	Trial 3	
0.00	0.384	0.390	0.241	0.338
5.00	15.1	13.5	9.6	12.7
10.00	18.7	17.8	12.2	16.2
20.00	17.8	20.6	21.8	20.1
30.00	21.8	20.6	27.8	23.4
45.00	16.3	70.1	20.6	18.5
60.00	27.8	23.0	-	25.4

Graph #3: *Variation of PGE_2 formation as a function of time*



Appendix D: Preliminary Experiment #2

Method:

1. Culture HEK-293 cells in two 12-well plates at 200,000 cells per well
2. Prepare and label Eppendorf microtubes to collect samples
3. Prepare 50.0 cm^3 of Hanks Buffer solution (See Hanks' Buffer Procedure)
4. Add 60.0 mm^3 of 10.0 $mmol.dm^{-3}$ arachidonic acid to form 6 cm^3 of 0.100 $mmol.dm^{-3}$ arachidonic acid intermediary solution.
5. Add the following volumes of 0.100 $mmol.dm^{-3}$ arachidonic acid solution into the corresponding Hanks' Buffer volumes to form 4 cm^3 solutions of different arachidonic acid concentrations:

Table #17: Specifics of serially diluted arachidonic acid solutions

Volume of 100 $\mu mol.dm^{-3}$ arachidonic acid (cm^3)	Volume of Hanks Buffer (cm^3)	Concentration of resulting solution ($\mu mol.dm^3$)
0.0400	4.00	1.00
0.200	3.80	5.00
0.400	3.60	10.0
0.800	3.20	20.0
1.20	2.80	30.0
2.00	2.00	50.0

6. Pipette 1.00 cm^3 of Hanks' Buffer into 3 wells for negative control.
7. Pipette 1.00 cm^3 of 1.00 $\mu mol.dm^3$ arachidonic acid in 3 wells each.
8. Repeat step 7 for subsequent concentrations of arachidonic acid (as found in step 5).
9. Incubate samples at 37°C for 30 minutes.
10. Remove samples from the incubator and collect in Eppendorf microtubes.

11. Freeze samples at -20°C for future enzyme immunoassay).

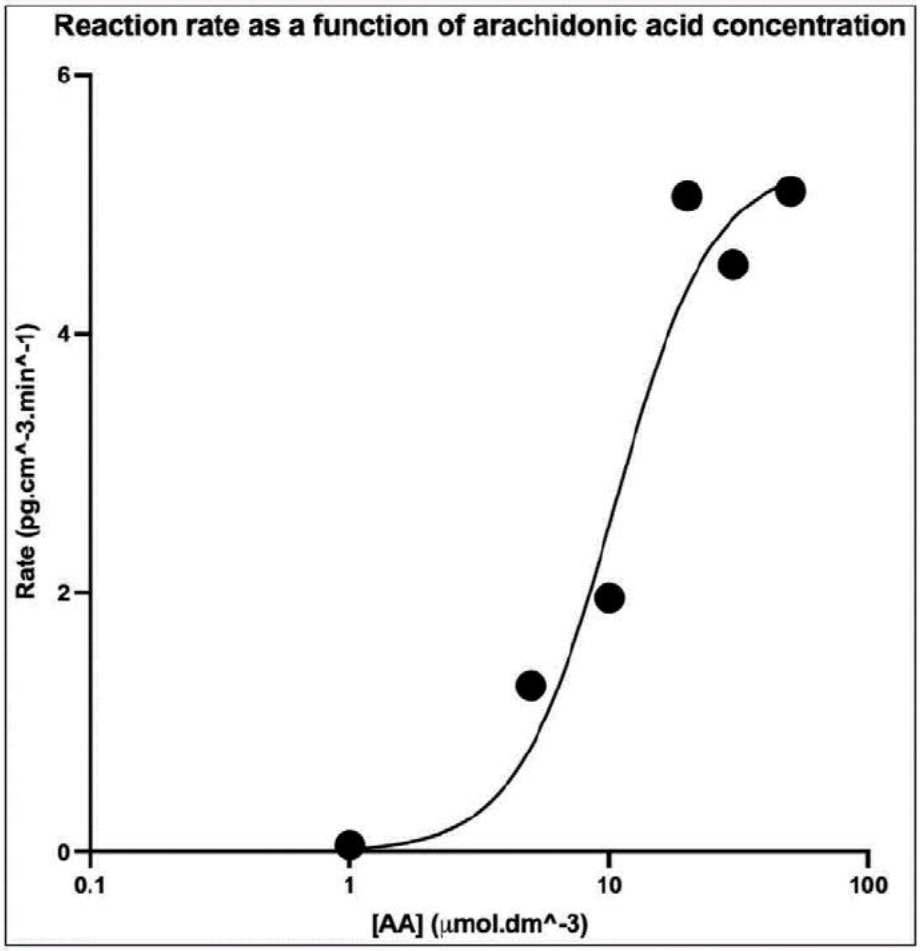
Table #18.1: Variation of PGE_2 concentration as a function of arachidonic acid concentration

[Arachidonic Acid] ($\mu mol.dm^{-3}$)	$[PGE_2]$ ($ng.cm^{-3}$)		
	Trial 1	Trial 2	Trial 3
0	0.826	0.607	1.812
1.00	1.42	0.775	2.32
5.00	39.1	37.6	18.4
10.0	56.4	60.9	98.2
20.0	145	159	417
30.0	121	148	139
50.0	152	155	152

Table #18.2: Variation of the PGE_2 rate of formation as a function of arachidonic acid concentration

[Arachidonic Acid] ($\mu mol.dm^{-3}$)	Rate ($ng.cm^{-3}.min^{-1}$)			Average Rate ($ng.cm^{-3}.min^{-1}$)
	Trial 1	Trial 2	Trial 3	
0	0.0275	0.0202	0.0604	0.02
1.00	0.0473	0.0258	0.0775	0.05
5.00	1.30	1.25	0.61	1.28
10.0	1.88	2.03	3.27	1.96
20.0	4.82	5.31	13.9	5.06
30.0	4.02	4.95	4.63	4.53
50.0	5.06	5.18	5.06	5.10

Graph #4: Variation of the PGE_2 rate of formation as a function of arachidonic acid concentration



Appendix E: Letter from External Laboratory Supervisor

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Beirut, September 27th, 2021

Dear ACS Extended Essay Coordinator,

This letter is to confirm and report on the lab supervision of Mr. Elias Mark Ziyadeh at my research laboratory at the Department of Biochemistry and Molecular Genetics at the faculty of Medicine at the American University of Beirut. The research project started on July 1st, 2021, and the lab work ended on September 15th, 2021.

After being introduced to our lab, its safety measurements, and the different equipment and techniques available, Elias independently further developed his aims and the experimental approaches after having previously discussed his hypothesis with me.

He has established each experimental approach of the different aims he defined, written in detail the experimental protocol, and submitted them to me for discussion. Elias showed good critical thinking with each step, discussing the feasibility of the approach, its limitations, and the number of replicates to use. He was trained on the different steps needed in the lab including using of micropipettes, filling 96 well plates, preparing buffers from stock solutions, and using sterile the cell culture hood techniques.

His experimental work was divided as follows:

1. Establishing the protocol to address to one aim at a time and writing it in detail. His protocols contained mainly 3 steps:
    a. Incubating the human cells overexpressing the enzyme of interest with the different reagents, which were available in the laboratory.
    b. Incubating the cells with the substrates and later inhibitors.
    c. Measuring the amount of prostaglandin E2 produced.
2. Collecting the data using the special software developed by our collaborators for the radio immunoassay/enzyme immunoassay.
3. Plotting the curves using Excel or Prism software and commenting on his results

1

4. Deciding on the next experimental step, interpreting all his results, and compare it to his hypothesis.

Elias used the enzyme immunoassay for prostaglandin E2 already established in the lab. This technique is based on very valuable immuno-measurements. Elias was able to learn and develop his own protocol for the dilution of the samples he obtained during his experiments, execute the measurement by filling the tiny wells, develop the colorimetric assay, calculate the data from the standard curve, and obtain the final results.

It was my pleasure to supervise Elias. During all his experiments, he expressed his strong will to do all the steps alone while being supervised but not replaced by any of us in the lab, identified mistakes in the experimentation, and was willing to repeat his experiments at times (admitting any experimental mistake is in my opinion highly ethical). Elias wrote all his protocols in detail, doing all the sophisticated calculations on his own and received feedback from time to time.

At all steps, Elias showed strong critical thinking, deciphering all steps, curve calculation, use of basal or standard controls, and the importance of vehicle control for the chemical reagents use (ethanol in his case). He also learned how to troubleshoot his results when faced with setbacks.

Finally, I would add that Elias is highly meticulous, extremely motivated, and an enthusiast. He is highly responsible, well organized, very keen on being ethically correct and independent. His performance during this study, both theoretically and in research, was above our expectations for university students, let alone a high school student.

Best regards,
Aida Habib, PhD



Supervisor comments:

***Supervisor**: By submitting this candidate work for assessment, you are taking responsibility for its authenticity. No piece of candidate work should be uploaded/submitted to the e-Coursework system if its authenticity is in doubt or if contradictory comments are added to this form. If your text in the box below raises any doubt on the authenticity of the work, this component will not be assessed.*

It was a great pleasure to supervise the extended essay of this candidate. He was totally engaged in the process from the very beginning. He chose a topic that interested him and was very excited to investigate about it. One of the most remarkable traits about this student is how independent he was. His EE was based on spectrophotometry. Though this technique was not practiced in the chemistry class, he was able to learn it very well. In addition to that, his work involved micro measurements which made it even harder. The procedure was long and complex. Several preliminary trials had to be done in order to design a procedure suitable to answer the research question. The candidate was always very excited to speak about his findings. He understood the purpose of every step of his experiment and was also good at using a new mathematical software. Despite the fact that he had to repeat the experiment several times, he never gave up. It is worth pointing out how much effort this student has put into his work whether on the research part or the experimental part. This experience helped him grow both on the academic and on the personal levels.

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